

Climate-Resilient Crop Advisory for Indian Farmers (MVP 2026)

Focus Region: North Indian Plains (Delhi, Punjab, Haryana, Western Uttar Pradesh)

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Guidelines Aligned: ICAR-NICRA Framework 2025-26, Vikaspedia Package of Practices, Soil Health Card Scheme 2026

Executive Summary

This advisory provides evidence-based, climate-adaptive agricultural guidance for the North Indian Plains, addressing critical vulnerabilities in **Wheat (Rabi), Rice (Kharif), and Mustard (Rabi)** production systems. The region faces escalating threats from **unseasonal temperature extremes, erratic precipitation patterns, and emerging pest pressures**. This document synthesizes ICAR research outputs and integrates actionable contingency measures to enhance farm-level climate resilience while maintaining productivity and profitability.

Section 1: Agronomic Calendar & Soil Matrix

1.1 Crop-Specific Agronomic Requirements

The following table consolidates critical agronomic parameters for the three principal crops of the North Indian Plains:

Parameter	Wheat (Rabi)	Rice (Kharif)	Mustard (Rabi)
Sowing Window	October 25 – November 30 (optimal: November 1-15)	May 15 – June 30	September 25 – October 31 (optimal: October 10-20)
Harvesting Window	April 15 – May 15	September 15 – October 30	January 20 – February 28
Crop Duration (days)	150-160	130-150 (direct seeding) / 150-160 (transplanted)	130-140
Critical Growth Stages	Tillering (30-40 DAS), Boot (80-90 DAS), Grain Filling (100-130 DAS)	Panicle Initiation (50-60 DAS), Heading (80-95 DAS), Flowering & Grain Maturation (100-140 DAS)	Flowering (35-50 DAS), Pod Development (55-85 DAS), Seed Maturation (90-130 DAS)
Optimal Temperature (Min °C)	7-12	20-25	10-15
Optimal Temperature (Max °C)	20-25	28-35	25-30
Soil pH Range	6.5-7.5	5.5-7.0	6.0-7.0

Parameter	Wheat (Rabi)	Rice (Kharif)	Mustard (Rabi)
Soil Texture (Preferred)	Loam to Sandy-Loam (well-drained, fertile)	Clay to Clay-Loam (high moisture retention)	Sandy-Loam to Loam (good drainage)
Organic Matter (%)	>2.5 (target: 3.0)	>2.0 (target: 2.5-3.0)	>2.0 (target: 2.5)
Nitrogen Requirement (kg/ha)	100-120 (split: 50 at sowing, 50 at tillering)	100-120 (60 kg basal + 40 kg top-dress at PI)	40-50 (applied at sowing, 25-30 kg)
Phosphorus Requirement (kg/ha)	50-60	40-60	40-50
Potassium Requirement (kg/ha)	30-40	30-40	20-30
Irrigation Schedule	4-6 irrigations: CRI (Crown Root Initiation, 21 DAS), Tillering (40 DAS), Boot (80 DAS), Grain Filling (100-120 DAS)	5-8 cm standing water throughout growing season; drainage at maturity	1-2 supplementary (if rainfall <40 mm during critical stages)
Water Requirement	400-500 (highly dependent on rainfall)	1000-1200 (includes rainfall + irrigation)	300-450

Parameter	Wheat (Rabi)	Rice (Kharif)	Mustard (Rabi)
Moisture (mm)			

1.2 Critical Soil Health Parameters (Soil Health Card 2026 Assessment)

Soil Health Card (SHC) Scheme Testing Protocol:

The Soil Health Card Scheme (implemented by Ministry of Agriculture & Farmers Welfare) assesses **12 critical soil parameters** for every farm holding, issued triennially:

Soil Parameter	Method & Unit	Desirable Range (North Plains)	Remedial Action if Deficient
Nitrogen (Available)	Alkaline KMnO ₄ method (kg/ha)	150-250	Add 25-50 kg N/ha through split application; use Azospirillum biofertilizer
Phosphorus (Available)	Bray's Phosphate method (kg/ha)	20-30	Apply Single Super Phosphate (SSP) @ 50 kg/ha or Rock Phosphate (RP) for long-term
Potassium (Available)	Ammonium Acetate method (kg/ha)	150-200	Apply Muriate of Potash (MOP) @ 30-40 kg/ha; combine with organic matter
Sulphur (Available)	Calcium Chloride extract (mg/kg)	10-20	Apply Gypsum @ 250-500 kg/ha for deficient soils; monitor in oilseeds (mustard)
Boron (DTPA extract)	Hot Water Soluble (mg/kg)	0.5-1.0	Apply Borax @ 10 kg/ha for mustard and wheat; critical for grain quality
Zinc (DTPA extract)	DTPA method (mg/kg)	1.0-1.5	Apply Zinc Sulphate @ 25 kg/ha or Zinc Oxide @ 10-15 kg/ha; foliar spray option
Iron (DTPA extract)	DTPA method (mg/kg)	4.0-8.0	Less common in North Plains; spray ferrous sulphate if chlorosis observed
Manganese (DTPA extract)	DTPA method (mg/kg)	2.0-5.0	Apply Manganese Sulphate @ 10 kg/ha if deficient (rare in North Plains)

Soil Parameter	Method & Unit	Desirable Range (North Plains)	Remedial Action if Deficient
Copper (DTPA extract)	DTPA method (mg/kg)	0.8-1.2	Apply Copper Sulphate @ 2 kg/ha; critical for fungal disease suppression
pH (Soil Acidity/Alkalinity)	Potentiometric method (1:2 soil:water)	6.5-7.5 (ideal neutral)	Acidic soils: apply lime @ 1-2 t/ha; Alkaline soils (high pH): incorporate S or FeSO _x
Electrical Conductivity (EC)	Conductivity meter (dS/m)	<1.0 (non-saline condition)	Saline soils: increase organic matter, apply gypsum, improve drainage
Organic Carbon (OC)	Walkley-Black method (%)	>2.0 target (currently 0.8-1.2 in North Plains)	Incorporate crop residues, apply 5 t/ha FYM or compost, adopt mulching

Soil Health Card Scheme Benefit (2026 Updates):

- **Coverage:** Over 25 crore Soil Health Cards distributed across India; ₹1706.18 crore released to states/UTs
- **Frequency:** Issued once every **three years** per farm holding
- **Digital Integration:** GIS-enabled portal, mobile app, QR-coded samples, geo-tagging for transparency
- **Farmer Access:** Free testing through government soil testing labs; recommendations aligned with crop requirements
- **Impact (2024-25):** 53 lakh new SHC cards issued; 1,987 village-level soil fertility maps prepared

Section 2: Biotic Stress Management (Pest & Disease)

2.1 Yellow Rust in Wheat (*Puccinia striiformis*)

2.1.1 Disease Identification & Symptoms

Causal Agent: *Puccinia striiformis* (fungal pathogen)

Key Identifying Features:

- **Appearance:** Long, narrow streaks of **yellow to orange pustules (uredia)** appearing along leaf veins on **upper leaf surfaces**
- **Progression:** Starts on lower leaves, spreads upward; pustules arranged in linear patterns following vascular tissue
- **Chlorosis:** Leaves turn yellow (chlorotic) with premature drying and reduced photosynthetic efficiency
- **Plant Impact:** Stunted growth if infected early (boot stage); significant grain filling losses in late infections
- **Sporulation:** Pustules contain millions of thin-walled urediniospores, spread by wind over long distances

Favorable Environmental Conditions:

Factor	Optimal Range	Duration Required
Temperat ure	10-15°C (optimum)	Growing phase: <23°C favors rapid multiplication
Humidity	>90% relative humidity	4-6 hours leaf wetness required for infection
Precipitat ion	Intermittent light rain or heavy dew	Cool, damp mornings accelerate spore germination
Crop Age	Late-sown wheat (November-December)	Susceptible from tillering through boot stage
Wind Condition s	10-20 km/h breeze	Facilitates spore dispersal over 100+ km

Critical Period in North India: December-February (during cool season); disease typically peaks in **late February-March** as temperatures warm.

2.1.2 Two-Tier Management Approach

Tier 1: Organic & Biopesticide Management

Strategy	Method & Application Rate	Timing & Frequency	Expected Efficacy
Varietal Resistance (Priority 1)	Sow rust-resistant varieties: PBW 343, PBW 502, HD 3118, WH 1105 (ICAR-recommended for North Plains)	Sowing time (October-November)	85-95% disease suppression if susceptible variety avoided
Crop Sanitation	Remove infected plant residues post-harvest; destroy volunteer wheat plants; avoid mixed cropping with wheat hosts	Post-harvest (May-June)	Reduces carryover inoculum by 60-70%
Crop Rotation	Rotate wheat with legumes, mustard, or maize; avoid 2+ consecutive wheat crops in same field	Every 2-3 years minimum	Breaks disease cycle; enhances soil nitrogen
Nitrogen Management	Apply nitrogen in split doses : 50 kg/ha at sowing, 50 kg/ha at tillering (avoid high N at late boot stage; excessive N increases disease severity)	Tillering & boot stage	Reduces disease by 25-30%; improves host resistance
Neem-Based Biopesticide (Barrix Control)	Spray Barrix Control (microbiostat) @ 0.3 g/liter water; acts as fungal growth inhibitor (non-systemic)	At first symptom appearance, repeat every 14 days	40-50% disease reduction; safe for pollinators
Sulphur Dusting (Preventive)	Dust wettable sulphur @ 25 kg/ha (early morning or late evening) starting at boot stage	First application at boot stage (80-90 DAS),	35-45% control; preventive rather than curative

Strategy	Method & Application Rate	Timing & Frequency	Expected Efficacy
		repeat 10-15 days later	
Field Monitoring & Scout Reports	Weekly field walks; record symptom appearance on bottom leaves using 0-9 scale (CIMMYT standard); use yellow sticky traps for disease forecasting	Weekly from December onward	Early detection saves cost and spray applications
Residue Management	Incorporate wheat straw (2-3 t/ha) post-harvest via disc harrow or rotavator to bury spore-bearing residue	Within 7 days of harvest (May-June)	Reduces primary inoculum source by 50-60% next season

Biopesticide Product Note: Barrix Control (Bacillus subtilis strain-based microbiostat) is approved by ICAR-NCIPM for organic farming; cost-effective alternative to chemical fungicides in early infection stages.

Tier 2: Recommended Chemical Management (Azole Fungicides)

Active Ingredient & Formulation	Spray Rate & Concentration	Timing of Application	Interval Between Sprays	Remarks
Hexaconazole 5% EC (DMI group)	1.0 L/ha in 500-600 L water (0.1% concentration)	First spray at boot stage (80-90 DAS) when first pustules appear; critical timing	14-day interval ; maximum 2 sprays/season	Cost-effective; broad-spectrum DMI fungicide; compatible with chlorothalonil
Propiconazole 25% EC	0.5-0.75 L/ha in 500-600 L water	At boot stage and repeat at grain-filling if symptoms persist	15-21 day interval	Good systemic translocation; local recommendation in Punjab/Haryana
Tebuconazole 25% EC	0.75-1.0 L/ha	Boot stage (critical); grain-filling stage (80-100 DAS)	10-14 days	Excellent systemicity; prevent resistance by alternating chemical families
Strobilurins (Optional, if combined with Septoria control)	Azoxystrobin 18.2% SC @ 0.75-1.0 L/ha	Boot & grain-filling stages	14-day interval	Do NOT apply if Septoria not present (cost optimization); excellent for dual-disease management

Chemical Management Decision Criteria:

- **Spray initiation:** Begin fungicide application when **first pustule** appears on bottom 2-3 leaves (Economic Threshold)
- **Maximum 2 sprays/season** to prevent resistance and minimize residue
- **Avoid spraying during grain-filling** if disease <5% severity (cost-benefit analysis)
- **Resistance management:** Alternate between DMI (azoles) and QoI (strobilurins) if 2+ applications required

Health & Safety Protocol:

- Spray early morning or late evening (avoid midday heat)
- Use ICAR-certified spray equipment (0.3 mm nozzle size)
- Farmers must wear full protective gear (mask, gloves, boots)
- Re-entry interval: 7 days post-spray
- Zero-residue harvest interval: 21 days minimum

2.2 Stem Borer in Rice (*Scirpophaga incertulas* – Rice Yellow Stem Borer)

2.2.1 Disease/Pest Identification & Symptoms

Causal Agent: *Scirpophaga incertulas* Walker (Lepidoptera: Crambidae) – primary stem borer species in North India; secondary pest: *Chilo suppressalis*

Life Cycle & Phenology:

Stage	Duration	Description	Damage Timing
Egg	7-12 days	Laid on leaf blade by female moth; hatches into neonates	Asymptomatic
Larva 1 (4 instars)	25-35 days	Bore into stem, feed on parenchyma tissue and vascular bundles; create tunnels; 4-5 mm creamy larvae	Active damage phase: 30-80 DAS (critical at PI)
Pupation	12-18 days (inside stem)	Pupate in stem cocoon; overwinter as diapausing larvae/pupae	Silent phase; no visible damage
Adult (Moth)	7-10 days (active)	Nocturnal; lay eggs on rice leaf blade; 25-30 mm wingspan, buff-colored wings	Egg-laying period: June-September (peak: July-August)

Generation in North India: 2-3 generations per season (depending on temperature); peak moth emergence: **mid-July to early August**

Visible Damage Symptoms:

- **Deadhearts (DH):** Yellowing and drying of central leaf (flag leaf stage) due to larval feeding in main stem; plant appears "burnt" at growing point
- **Whiteheads (WH):** Unfilled or poorly-filled panicles resulting from larval damage during panicle initiation; entire panicle turns white/silvery
- **Stem Tunneling:** Visible holes or slits on internodes (30-60 cm height); brown frass (larval excretion) at entry points
- **Grain Loss:** Yield reduction of 25-40% if severe infection during flowering/grain-filling
- **Ripeness & Breakage:** Weakened stem prone to lodging (falling) under rainfall or wind

Economic Threshold: 5-10% **deadhearts at active tillering stage** or 10-15% **whiteheads at panicle initiation** triggers intervention

2.2.2 Two-Tier Management Approach

Tier 1: Organic & Biopesticide Management

Strategy	Method & Application Rate	Critical Timing	Expected Efficacy
Varietal Resistance (Primary)	Cultivate resistant varieties: IR 36, ADT 47, Saryu, Sugandhya (ICAR-recommended, high-yield + stem-borer tolerance)	Sowing/transplanting (May-June)	70-85% suppression of larval population
Cultural Control - Field Sanitation	Destroy/mulch rice stubble by June-end; plow field immediately post-harvest (February-March) to expose pupae to sun/predators	Post-harvest (February-March)	Destroys 50-60% of overwintering pupae
Synchronized Crop Planting	Plant rice in compact 2-3 week blocks across village (community approach) to disrupt moth emergence patterns and reduce egg-laying period	Transplanting: May 15-June 15 (block planting)	Reduces peak infestation by 30-40%
Biological Control - Predator y Wasps (Trichogramma)	Release Trichogramma chilonis (parasitoid wasp) @ 50,000-100,000 wasps/ha in 2-3 releases at 10-day intervals starting from June 15	Egg-laying peak (mid-June to early July)	50-70% egg parasitization; breaks reproductive cycle
Neem-Based Biopesticide (Organeem)	Spray Organeem (neem extract) @ 3% solution (30 ml/liter water) at tillering & boot stages (25-35 DAS and 50-60 DAS); acts as insect growth regulator	25-35 DAS (active tillering), repeat at 50-60 DAS	40-60% larval suppression; safe for predators
Bacillus thuringie	Apply Larvo Raze (Bt-based bio-larvicide) @	40-50 DAS when	50-65% larval mortality if

Strategy	Method & Application Rate	Critical Timing	Expected Efficacy
nsis (Bt) Biopesticide	2.5 ml/liter water; targets 2nd-4th instar larvae (cryptic stage)	larvae entering stems (before deadhearts)	applied before stem entry
Pheromone Trap Monitoring	Deploy pheromone traps (sex pheromone: Z-11-hexadecenal) @ 4-5 traps/acre; monitor moth catches daily	Start traps by mid-June; monitor thrice weekly	Forecasts peak egg-laying; alerts farmers to spray timing; 25-35% adult moth capture
Fertilizer & Irrigation Timing	Apply nitrogen in split doses to avoid lush vegetative growth (reduces host attractiveness); maintain 5-7 cm water level (avoid excess water during tillering)	Basal (20 kg N) + 40 kg N at PI stage	Balanced nutrition reduces susceptibility by 15-20%

Biopesticide Product Availability in North India:

- **Trichogramma chilonis:** Produced by ICAR-IARI (New Delhi) and NCDS; cost ₹200-250/100,000 wasps; distribute via ATMA/KVK
- **Larvo Raze & Organeem:** Private companies (Kay Bee Bio, others) supply; cost ₹400-600/liter
- **Pheromone Traps:** ₹30-50/trap; reusable with pheromone vials (lasts 4-6 weeks)

Tier 2: Recommended Chemical Management (Insecticides)

Active Ingredient & Formulation	Spray/Application Rate	Critical Timing	Interval & Repeats	Remarks
Triflumezopyrim 10 SC (ICAR-field proven)	100-125 ml/ha in 500-600 L water (10% of recommended dose often effective)	40-50 DAS (pre-stem entry; before deadhearts appear)	Single spray; if infestation persists, repeat at 60-70 DAS	Highest efficacy reported by IARI/ICAR (2024-25 trials); systemic action; low mammalian toxicity
Flupyrim 20 SC	150-200 ml/ha in 500-600 L water	40-50 DAS (tiller stage; critical window)	10-15 day interval if required	New-generation insecticide; effective at lower doses; compatible with biopesticides
Profenophos 50% EC (Traditional, budget option)	1.0 L/ha in 500-600 L water	At 40-50 DAS & 60-70 DAS (2 sprays if needed)	14-21 day interval	Broad-spectrum organophosphate; cost-effective (₹150-200/L) but higher toxicity; requires PPE
Carbofuran 3% Granules (Soil Application, alternative)	8-10 kg/ha mixed in soil at sowing/transplanting	At transplanting/sowing time	Once during season; combined with spray if needed	Systemic action through roots; persistent protection (30-40 days); less water pollution

Spray Protocol & Economic Threshold Determination:

Symptom	Action	Timing
Pheromone trap catches >5-8 moths/trap/night	Start first spray within 48 hours	40-50 DAS (before egg-laying peak)
5-10% deadhearts visible	Apply first chemical spray immediately	50-60 DAS (active invasion)
10-15% whiteheads at PI	Spray at panicle initiation (55-65 DAS)	Before heading (80-90 DAS)
No traps or monitoring available	Prophylactic spray at 40-50 & 60-70 DAS	Default schedule for risk-averse farmers

Health & Safety (Insecticide Application):

- **Spray application:** Early morning (5-7 AM) or late evening (5-7 PM); avoid midday/windy conditions
- **Re-entry interval:** 7 days minimum (Profenophos: 10 days)
- **Harvest interval:** 14 days minimum post-spray
- **Personal protective equipment (PPE):** Mask, gloves, boots, full sleeve shirt mandatory
- **Waste disposal:** Empty pesticide containers buried in designated pit away from water source

2.3 Aphids & Whitefly in Mustard (*Lipaphis erysimi* – Mustard Aphid; *Bemisia tabaci* – Whitefly)

2.3.1 Pest Identification & Symptoms

Mustard Aphid – *Lipaphis erysimi* Kaltenbach (Hemiptera: Aphididae)

Feature	Description
Morphology	Small, 1.5-2.5 mm, dark yellow to pale yellow body; antennae long, 6-segmented; wingless (apterous) & winged (alate) forms present
Life Cycle	8-12 day generation time (at 25-30°C); 8-10 generations/season in North India
Seasonal Occurrence	Peaks: December-January (peak infestation stage) & January-February (pre-flowering to seed fill)
Host Preference	Highly specific to Brassica crops (mustard, rapeseed, cabbage, cauliflower); polyphagous on 100+ plant species at lower preference

Whitefly – *Bemisia tabaci* (Hemiptera: Aleyrodidae)

Feature	Description
Morphology	Tiny, 1-2 mm, white body, yellow eyes; nymphs flat, greenish; both sexes winged
Seasonal Occurrence	Secondary pest in mustard; co-infests with aphids in January-February
Vector Role	Transmits plant viruses (Yellow Vein Mosaic Virus - YVMV); damages via sooty mold deposition

Damage Symptoms (Joint Aphid-Whitefly Infestation):

Symptom	Timing	Economic Impact
Leaf Yellowing & Curling	Flowering to pod development (35-60 DAS)	Reduced photosynthesis; <5-10% loss if <200 aphids/plant
Stunted Pod Development	Pod fill stage (55-85 DAS)	Pod shattering; reduced seed set; yield loss 30-40% if unchecked
Honeydew Secretion & Sooty Mold	Late season (80+ DAS)	Black fungal coating blocks light; secondary fungal infections (brown spot, white rust)
Virus Infection (YVMV)	If whitefly present	Causes vein yellowing; severe necrosis; >50% yield loss in susceptible varieties
Seed Quality Deterioration	Pre-harvest (110-130 DAS)	Oil content reduced by 3-5%; seed vigor impaired

Economic Threshold for Intervention:

- **Aphids: 100-150 aphids per plant** (flowering to early pod stage); **200-250/plant** (late pod stage tolerance higher)
- **Whitefly: 5-10 insects per leaf** (threshold lower due to virus-transmission risk)
- **Combined:** If both pests co-exist, **lower threshold applied** (spray at 80-100 aphids/plant + whitefly presence)

2.3.2 Two-Tier Management Approach

Tier 1: Organic & Biopesticide Management

Strategy	Method & Application Rate	Critical Timing	Expected Efficacy
Varietal Resistance (Priority 1)	Select tolerant/resistant varieties: Pusa Mustard 27, PM 30, RH 0406, NRCHB 101 (IARI-bred; adverse effects on aphid host preference & nymphal survival)	Sowing (September-October)	25-30% natural resistance; reduces pest buildup by mid-season
Neem-Based Biopesticide (NSKE 5% or Organeem 3%)	Neem Seed Kernel Extract (NSKE) @ 5% = 50 ml neem extract concentrate/liter water; or Organeem 3% concentrate @ 30 ml/liter ; apply as foliar spray	Flowering stage (35-45 DAS) & repeat at pod development (55-65 DAS) ; 2-3 sprays for severe infestation	40-60% reduction in aphid & whitefly numbers; acts as insect growth regulator (disrupts reproduction)
Neem Oil (1%) + Fish Oil Resin Soap (2.5%)	Mix 1 liter neem oil + 2.5 kg fish oil resin soap in 100 liters water; spray thoroughly (wetting underside of leaves)	Weekly spray starting at flower bud stage (30-35 DAS) through early pod fill (70 DAS)	50-70% pest suppression; physical smothering + insecticidal action; safe for beneficial insects
Soap Spray (Insecticidal Soap)	Potassium soap @ 2.5-3% concentration (25-30 g/liter); or dish soap @ 1% dilution (emergency measure)	Weekly or bi-weekly during peak infestation (December-January)	30-40% control; effective against soft-bodied insects; non-persistent
Yellow Sticky	Deploy bright yellow sticky cards @ 4-5	Install at flowering (35-	15-25% adult capture; early

Strategy	Method & Application Rate	Critical Timing	Expected Efficacy
Traps	cards/acre at plant height; attracts aphids & whiteflies	40 DAS) through pod fill (80 DAS)	warning signal for population build-up; non-toxic monitoring tool
Botanical Pesticides (Pyrethrin Extract)	Pyrethrum spray @ 0.05% concentration (derived from Chrysanthemum flowers)	At peak infestation (January-February)	35-50% efficacy; rapid knockdown; safe for vertebrates; non-persistent (repeat every 5-7 days)
Companion Planting & Pushpa Mulch	Plant garlic/onion rows in mustard field (1 row per 3 mustard rows); or apply mustard straw mulch (3-4 t/ha) post-sowing	Sowing time (September-October) & maintenance (November-December)	10-20% natural suppression via repellent volatiles; improves soil organic matter
Biological Control - Predatory Lady Beetles	Encourage/release Coccinella septempunctata or Menochilus sexmaculatus (lady beetles); consume 50-100 aphids/day per beetle; naturally prevalent in North India	February-March (post-winter) emergence; encourage via minimal pesticide use	30-40% aphid suppression via natural predation
Field Sanitation &	Remove infested leaves/plant parts; destroy crop residue	Post-harvest (March-April); pre-sowing	Breaks pest life-cycle; reduces

Strategy	Method & Application Rate	Critical Timing	Expected Efficacy
Residue Removal	immediately post-harvest; avoid mixed farming with other Brassica crops	field preparation (August-September)	carryover population by 50-60%

Biopesticide Product Recommendations (Cost-Benefit):

Product	Cost/liter	Efficacy	Farmer Access (North India)
NSKE 5% (Home-made)	₹0 (DIY)	40-50%	Teach via ATMA; easy preparation from neem seeds
Organeem (Commercial)	₹300-400/liter	50-60%	Available via dealers; ICAR-approved
Neem Oil + Soap	₹500-800 (total cost/season)	50-70%	Easily available; economical
Pyrethrum Extract	₹800-1200/liter	35-50%	Premium product; for severe cases only

Tier 2: Recommended Chemical Management (Insecticides)

Active Ingredient & Formulation	Spray Concentration & Rate	Critical Timing	Spray Interval	Key Remarks
Thiamethoxam 25 WG (Neonicotinoid class)	0.2 g/liter water (0.005% dilution); = 100-125 ml of 10% solution/ha	Flowering (35-45 DAS) - first spray; Early pod stage (55-65 DAS) - if population rebounds	10-14 day interval if required; maximum 2 applications/season	BEST EFFICACY per IARI field trials (2024-25); lowest 6.81 aphids/plant; yield 1701.96 kg/ha (BCR 2.47:1)
Dinotefuran 20 SG (Neonicotinoid)	0.25 g/liter water = 125 ml of 20% solution/ha	40-45 DAS (early flowering); repeat at 60-70 DAS	12-15 day interval	Second-best option; 10.62 aphids/plant post-spray; yield 1628.62 kg/ha; lower mammalian toxicity
Imidacloprid 17.8 SL (Neonicotinoid, budget option)	0.15 ml/liter water = 75-100 ml/ha	Flowering to early pod stage (40-60 DAS)	12-14 day interval; 1-2 applications	Cost-effective; systemic action; effective against whitefly; residue concern in export-quality mustard

Active Ingredient & Formulation	Spray Concentration & Rate	Critical Timing	Spray Interval	Key Remarks
Pyrethroid + Neonicotinoid (Combination)	Cypermethrin 10% EC @ 0.05% + Imidacloprid 17.8 SL @ 0.15 ml/liter	Dual spray at 40-50 DAS & 60-70 DAS	Alternate chemicals to manage resistance	Broadens pest control spectrum; effective against co-pests

Decision Criteria for Chemical Spray Initiation:

Condition	Action
100-150 aphids/plant at flowering	Start first spray immediately (Thiamethoxam preferred)
>200 aphids/plant at pod fill + whitefly presence	Spray urgently ; consider neonicotinoid (high efficacy against both)
<100 aphids/plant, no whitefly, no virus symptoms	Monitor weekly ; defer spray; use organic control (NSKE, neem oil) as alternative
Organic farming certification	Neem oil, NSKE, pyrethrum only ; yellow traps + predatory beetles supplementary

Spray Application Protocol:

- **Nozzle type:** 0.4-0.5 mm nozzle (flat fan); ensure thorough coverage of leaf undersides (where aphids congregate)
- **Spray volume:** 500-600 L/ha (ground application); 50-75 L/ha if aerial/drone spray
- **Timing:** **Early morning (5-7 AM)** or **late evening (5-7 PM)** to avoid bee activity and maximize leaf coverage
- **Phytotoxicity check:** Spray 2-3 plants first; wait 24 hours to ensure no leaf burn before mass application
- **Re-entry interval:** 3-5 days (neonicotinoids); 7 days (pyrethroids)
- **Harvest interval:** **21 days minimum** post-spray to ensure residue compliance (especially for export markets)

Environmental & Health Safeguards:

- **Pollinator protection:** Spray when bees inactive (early morning/evening); avoid in-bloom application
 - **Groundwater safety:** Do NOT apply neonicotinoids near wells or water sources (risk of leaching)
 - **Worker safety:** Mandatory PPE (gloves, mask, boots, full-sleeve shirt); restrict entry post-spray
 - **Waste management:** Triple-rinse empty pesticide bottles; bury in designated pit away from habitation
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Section 3: Climate Contingency Plan (Extreme Weather Events)

3.1 Terminal Heatwave Response (February-March Grain-Filling Crisis)

Climate Scenario: Unseasonal heat spike during **critical grain-filling stage (100-130 DAS)** causing:

- Temperature surge to **30-35°C** (normal: 20-25°C)
- Low relative humidity (<30%)
- Accelerated crop maturity; reduced grain weight by 15-30%
- Increased water stress; wilting symptoms

Affected Crops & Vulnerability:

- **Wheat:** Boot to milky grain stage (90-110 DAS) – most vulnerable
- **Rice:** Post-heading to dough stage (100-140 DAS) – moderate vulnerability
- **Mustard:** Seed maturation (100-130 DAS) – pod abortion risk

3.1.1 Immediate Emergency Actions (First 24-48 Hours)

Priority	Action	Implementation Method	Expected Benefit
Action 1: Emergency Irrigation	Provide supplementary irrigation at 40-50 mm depth immediately upon heatwave alert; prevent soil moisture stress	Flood irrigation (furrow method) if available; drip/sprinkler if equipped	Cools soil; maintains turgor; reduces heat stress by 3-5°C at leaf surface
Action 2: Spray Cool-Water Mist (Wheat/Mustard only)	Apply fine water mist via sprinkler at evening (5-7 PM) to wet foliage; evaporative cooling mechanism	Mobile sprinkler/overhead system; 30-45 minutes spray duration	Reduces canopy temperature by 2-4°C; improves grain filling rate by 10-15%
Action 3: Anti-Transpirant Foliar Spray	Spray Kaolin clay suspension (5-7%) or organic film-forming anti-transpirants @ 5 L/ha to reduce water loss	Mix kaolin with water to form milk-like emulsion; spray in evening	Reduces transpirational water loss by 30-40%; maintains leaf turgor for 5-7 days
Action 4: Nutrient Boost via Foliar Application	Apply Urea 2% solution (20 g urea/liter water) or DAP 1% solution (10 g DAP/liter) via foliar spray in evening	Dissolve fertilizer in water; apply via knapsack sprayer covering all leaves	Compensates for reduced nutrient uptake due to heat stress; supports grain filling physiology

3.1.2 Short-Term Mitigation (Week 1 Post-Heatwave)

Strategy	Timing & Application	Operational Details	Impact
Increased Irrigation Frequency	2nd irrigation within 5-7 days of heatwave onset (instead of standard 10-15 day interval)	Apply 40-50 mm per irrigation; monitor soil moisture via tensiometer (ideal: 15-20 kPa tension)	Maintains soil water availability; mitigates yield loss by 5-10%
Mulching (Mustard/Late-Sown Wheat)	Apply crop residue/straw mulch (5-10 cm depth) around plant base within 2-3 days of heatwave	Use locally available crop residue (rice straw, wheat straw); avoid mulch contact with stem (prevent rot)	Reduces soil temperature by 2-3°C; conserves soil moisture; reduces evaporative loss by 25-30%
Intercropping Shade Management	If intercropped (wheat-mustard or sugarcane-wheat systems), DO NOT prune adjacent taller crops during heat; maintain natural shade	Maintain canopy cover; shade reduces direct solar radiation on primary crop by 20-30%	Reduces leaf temperature by 3-5°C; minimizes heat-stress symptom progression
Harvest Timing Decision	Monitor maturity daily via grain dough-stage visual assessment ; harvest 2-3 days earlier than normal if heatwave continues	Check 10-15 heads across field; harvest when grains reach wax/soft-dough stage (moisture 16-18%) to avoid over-maturity	Prevents excessive drying; retains grain weight; reduces shattering loss

3.1.3 Post-Heatwave Recovery & Damage Assessment (Week 2+)

Activity	Assessment Criteria	Decision Framework
Yield Loss Estimation	Count 20-25 spots per field (0.5m × 0.5m); count viable grains per head; compare to pre-heatwave phenology	If grain weight <85% of expected (e.g., 35 g/1000 grains vs. 40-45 normal), loss confirmed; assess for insurance/government relief eligibility
Insurance Claim Documentation	Photograph affected field areas showing wilting, grain abortion, stunted growth; obtain government enumerator's report (via revenue dept.)	Threshold for claim: >20% yield loss; required documents: land records, insurance policy, crop-cutting report
Soil Recovery Actions	Post-harvest: Apply FYM/compost @ 5-10 t/ha to replenish organic matter depleted via heat stress; incorporate crop residue	Improves soil water-holding capacity by 15-20%; restores microbial community

3.2 Unseasonal Heavy Rain & Hail During Harvest (September-October for Rice; April-May for Wheat)

Climate Scenario: Unexpected **precipitation (80-120 mm)** or **hail storm** during grain maturation/harvest causing:

- **Lodging** (crop falling/bending); grain quality deterioration
- **Grain sprouting** (pre-harvest germination); reduced milling quality
- **Disease spike** (fungal infections via leaf wetness)
- **Mechanical loss** (shattering, mud-caking of grain)

Most Vulnerable Stages:

- **Rice:** Dough stage (130-140 DAS) onward – grain absorbs moisture; sprouting risk
- **Wheat:** Milk to dough stage (100-120 DAS); late-sown (November) crops at maturity (April-May) coinciding with pre-monsoon rains
- **Mustard:** Seed maturation (110-130 DAS); high moisture causes pod rotting, oil rancidity

3.2.1 Immediate Emergency Actions (First 24-48 Hours After Heavy Rain/Hail)

Priorit y	Action	Implementation Method	Expected Benefit
Action 1: Assess Lodging Severity	Inspect 20-25 spots across field; note % of plants standing vs. fallen; document with photographs for insurance/relief	Walk field systematically; use smartphone to geo-tag affected areas; report to gram panchayat within 24 hours for official assessment	Establishes baseline for compensation claim; guides harvesting strategy
Action 2: De- watering / Drain Standi ng Water	If hail + heavy rain causes waterlogging: open furrows/channels to drain excess water within 6-8 hours ; prevent root-zone saturation	Use spade/hoe to breach bunds; create diagonal drainage channels; ensure 2-3% field slope for gravity drainage	Prevents anaerobic conditions; avoids fungal root-rot; reduces grain sprouting risk by 50-60%
Action 3: Diseas e Surveil lance & Preven tive Spray	Scout field for leaf spots, panicle blast (rice), or sclerotinia rot (mustard) ; spray Mancozeb 75% WP @ 1.5 kg/100 L water within 24 hours of rain	Inspect 10-15 plants across field; focus on leaf blade, panicle, and basal stem for lesions	Prevents secondary fungal epidemic; reduces yield loss due to disease by 10-15%

Priorit y	Action	Implementation Method	Expected Benefit
Action 4: Readin ess for Emerge ncy Harves t	Mobilize labor & harvesting equipment (combine/manual sickles) within 48 hours of rain event; prepare threshing floor; arrange tarpaulins	Contact 15-20 farm workers (if manual harvest) or book combine harvester (if available in village); ensure threshing floor is dry/covered	Minimizes shattering loss (2-3% per day of delay); prevents grain sprouting (each day of high moisture = 1-2% quality loss)

3.2.2 Short-Term Mitigation (Week 1 Post-Event)

Strategy	Operational Details	Timing	Impact
Accelerated Harvest (Lodged Crops)	For lodged rice/wheat: switch to manual harvesting (sickle) above lodged grain rather than combine (reduces grain loss via trampling); collect fallen grain separately	Manual harvest 2-3 days earlier than normal; handle lodged grain carefully; separate grain into high-quality (upright) & lower-quality (lodged) lots	Prevents additional loss from decomposition, disease, or animal damage; salvages 60-70% of lodged grain value
Increased Drying Efforts	Spread harvested grain on clean, elevated threshing floor (not directly on ground) ; use tarpaulins during rain , open for sun-drying; target moisture <14% (from post-rain ~20-25%)**	Arrange grain in 2-3 cm layers; turn every 2-3 hours; use mechanical dryer if available (capacity-dependent)	Reduces grain sprouting by 80-90%; restores milling quality from 60% to 85-90%
Quality Assessment & Segregation	Post-drying: test 1-2 kg sample grain for germination % (soak & observe for radicle protrusion) ; segregate sprouted grain (off-grade, lower price)	Sprouted grain: sell separately at 20-30% discount; use for animal feed, seed production avoided	Maintains higher-grade grain for market sale; minimizes economic loss via proactive grading
Storage Management	Post-harvest storage: ensure grain moisture 12-13% (use moisture meter); store in airtight containers or godowns with lime/silica gel desiccants ; avoid bins	Lime @ 2% wt of grain absorbs moisture; silica gel: 5 kg per ton of grain	Prevents mold growth, maintains storability for 6-9 months

Strategy	Operational Details	Timing	Impact
	with rainwater ingress		

3.2.3 Post-Event Recovery (Week 2+)

Activity	Details	Documentation for Relief
Crop Insurance Claim Filing	Pradhan Mantri Fasal Bima Yojana (PMFBY) threshold: >20% yield loss + photographic evidence + official crop-cutting report from taluka office	Submit within 72 hours of rain event; contact insurance agent with farm documentation (land record, policy number); ensure village revenue official files preliminary report
Government Relief Application	Apply to District Collector's office under Disaster Management Act if losses exceed 33% at revenue village level (village-level assessment required)	Provide: affected land area, crop type, % loss estimated, bank account (for Direct Benefit Transfer); aggregate village loss assessed by taluka officer
Soil Recovery Post-Harvest	After harvest: incorporate crop residue + apply FYM (3-5 t/ha) to restore soil structure damaged by waterlogging; improve drainage via sub-soiling if available	Residue incorporation prevents erosion; organic matter restores soil aeration; prepares field for next crop

Section 4: Policy & Financial Support for Farmers (2026 Updates)

4.1 PM-KISAN Scheme 2026 Updates

Pradhan Mantri Kisan Samman Nidhi (PM-KISAN): Direct income support to farmers

Parameter	Details (2026 Update)
Beneficiary Eligibility	All land-holding farmers (marginal, small, medium) irrespective of land size; excluded: high-income individuals (income tax assessee) & government employees
Benefit Structure	₹6,000/year (₹2,000 per installment × 3 installments) paid directly to registered bank account; installments: February, May, August
Digital Integration (2026)	Biometric authentication for installment disbursement; geo-tagged field verification via mobile app to prevent duplicate registration; real-time dashboard for status tracking
Recent Changes (2025-26)	Expanded coverage to tenant farmers & agricultural laborers (pilot states); inclusion of female co-owners as joint beneficiaries (2-3 women per household priority)
Disbursement Status (2025)	Over 11.5 crore farmer accounts registered; ₹1.41 lakh crore disbursed cumulatively; 90%+ success rate in North India (Delhi, Punjab, Haryana, UP)
Application Process	Online: pmkisan.gov.in or UIDAI portal; offline: revenue office (gram panchayat/taluka) with Aadhaar, bank account, land records
Non-Beneficiary Redressal	If eligible farmer excluded: appeal to Revenue Circle Officer (taluka level) with land proof; correction turnaround: 7-15 days

PM-KISAN Integration with Climate Resilience:

- **Linkage to crop insurance:** PM-KISAN beneficiary status now **prerequisite for PMFBY enrollment** (simplified); reduces documentation burden
 - **Green farming incentive (proposed 2026):** ₹500 additional annual support for farmers **adopting soil health card recommendations** (verified via SHC database)
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4.2 Soil Health Card (SHC) Scheme 2026 Benefits & Digital Enhancement

Objective: Provide farmer-specific, science-based soil health data & nutrient recommendations; promote precision fertilizer use

4.2.1 SHC Scheme Mechanics (Triennial Cycle)

Phase	Timeline	Process & Farmer Role
Soil Sampling	Once every 3 years	Government/contracted sampling team collects 5-10 soil cores from farmer's field; geo-tagged via GPS; QR-coded sample vial
Laboratory Testing	15-21 days	12 parameters tested: N, P, K, S, B, Zn, Fe, Mn, Cu, pH, EC, OC (Organic Carbon)
Report Generation	21-30 days	Soil Health Card issued (printed + digital via SMS/WhatsApp); includes nutrient status & crop-specific recommendations
Fertilizer Recommendation	Post-card receipt	Custom recommendation based on tested soil values (e.g., if P = 15 kg/ha, recommend SSP application; if OC = 0.8%, recommend FYM 5 t/ha)
Farmer Application & Verification	3-month application window	Farmer applies recommendations; optional follow-up verification by KVK/ATMA to assess yield impact

4.2.2 SHC Scheme Benefits (Quantified Impact)

Benefit Category	Impact Evidence (2024-25 ICAR Data)
Increased Productivity	Farmers following SHC recommendations achieved average yield increase of 8-15% compared to control (e.g., wheat: 45 q/ha vs. 40 q/ha; rice: 55 q/ha vs. 50 q/ha)
Reduced Input Cost	Balanced fertilizer application reduced NPK expenditure by 15-20% ; avoided excess urea application (common in North India) saving ₹1,500-2,500/ha
Soil Health Improvement	Farmers with 2+ consecutive SHC cycles showed improved OC from 0.8% to 1.2-1.5% (20+ months); EC maintained <1.0 dS/m (non-saline condition)
Sustainability Metrics	Reduced chemical fertilizer dependency; environmental benefit: 20-30% lower nitrogen runoff to groundwater (measured in Punjab pilot)
Farmer Satisfaction	Survey data (2024-25): 87% farmers reported SHC recommendations "useful/very useful" for decision-making; 64% adopted recommendations fully

4.2.3 SHC Digital Innovations (2026 Rollout)

Innovation	Feature & Farmer Benefit
GIS-Enabled SHC Portal	Farmers access digital map of village soil fertility status; compare their field with neighboring fields; identify nutrient hotspots for targeted intervention
Mobile App (SHC Mobile)	Real-time soil sample collection scheduling ; geo-tagged photo uploads for verification; instant SMS notification upon report generation; QR-code scanning for report access
AI-Powered Crop Advisory	Machine learning algorithm integrates SHC data + weather forecast + crop stage to generate personalized fertilizer/irrigation timings (pilot: Delhi, Haryana)
Direct B2B Marketplace (Proposed)	Soil test results linked to fertilizer recommendations visible to agrochemical dealers; farmers get bulk-order discounts on recommended products (reduces intermediaries)

Cost-Benefit of SHC Adoption:

Cost	Benefit	Net ROI
Free soil testing (government subsidy)	Yield increase: 8-15% (value: ₹3,000-5,000/ha at current prices)	Benefit-Cost Ratio: 1:4-6 (highly profitable)
Fertilizer optimization cost: ₹500-800 (farmer-borne for implementation)	Input cost savings: ₹1,500-2,500/ha	ROI: 200-300% within single season

4.2.4 Accessing SHC Services (2026 Process - North India)

Step-by-Step Procedure:

- 1. Registration:** Contact **village-level soil sampling team** (via gram panchayat/ATMA office) or **state SHC portal** (<https://soilhealth.dac.gov.in>); provide Aadhaar & land record
- 2. Field Sampling:** Scheduled within 15-30 days; team collects 5-10 soil cores (top 15 cm) from representative field areas
- 3. Lab Testing:** Sample processed at **state/district soil testing laboratory**; turnaround 15-21 days

4. Report Retrieval:

- **Option A (Digital):** SMS/WhatsApp notification with report; download via QR-code scan or mobile app
- **Option B (Print):** Collect physical card from taluka office/gram panchayat

5. Recommendation Application: Fertilizer recommendations printed on card; farmer implements with support of **village-level extension worker (VEW)** or **Krishi Gyan Kendras (KGKs)****Contact Details (North India SHC Nodal Offices):**

State	Nodal Ministry	Portal Access	Farmer Helpline
Delhi	Dept. of Agriculture (NDSE)	soilhealth.dac.gov.in	011-XXXX-XXXX (toll-free: 1800-XXX)
Punjab	Punjab Agriculture Department	https://soilhealth.punjab.gov.in	0172-XXXX-XXXX (24/7 helpline)
Haryana	Haryana Department of Agriculture	https://haryana.gov.in/soil-health	0172-XXXX-XXXX
Uttar Pradesh	UP Department of Agriculture	https://upagriculture.com	0542-XXXX-XXXX

4.3 Complementary Government Schemes & Linkages (2026)**Integrated Support System for Climate-Resilient Farming:**

Scheme	Target Benefit	Linkage to SHC/PM-KISAN	Access
Pradhan Mantri Fasal Bima Yojana (PMFBY)	Crop insurance covering yield losses (>20%) due to adverse weather, pests, diseases	Auto-enrollment for PM-KISAN beneficiaries; premium: 2% (kharif), 1.5% (rabi); government subsidy: 88%	soilhealth.dac.gov.in (linked portal)
National Mission for Sustainable Agriculture (NMSA)	Per Drop More Crop (PDMC) micro-irrigation subsidy; Rainfed Area Development	SHC findings inform irrigation recommendations (e.g., low-permeability soils → drip system priority)	pradhanmantriyojana.gov.in
Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)	Irrigation infrastructure development; micro-irrigation subsidies (50-90% cost-share)	SHC identifies water-stressed fields for priority allocation	pradhanmantriyojana.gov.in
Aatma Nirbhar Bharat Package (Agricultural Component)	Concessional credit (2% interest) for farm mechanization, micro-irrigation, organic farming transition	PM-KISAN beneficiary status & SHC compliance accelerates loan approval	Cooperative banks, NABARD district offices
Rashtriya Gokul Mission	Support for fodder cultivation & dairy co-operatives; complements crop diversification	Encourages farmers to adopt mustard + fodder intercropping systems	dairy.gov.in

Section 5: Integrated Recommendations & District-Specific Advisories

5.1 Recommended Cropping Systems for North Plains (Climate-Resilient Pattern)

Based on soil type, water availability, and climate vulnerability:

Cropping System	Suitability by District	Rotation Pattern	Climate Resilience Feature
Wheat-Rice-Mustard (Sequential/ Triple-Crop System)	High-rainfall zones (Punjab, Haryana East)	Wheat (Oct-May) → Rice (May-Oct) → Mustard (Oct-Feb) overlap	Mustard provides winter income diversification; reduces rice dependency
Wheat-Mustard (Dual Crop)	Medium-rainfall, water-scarce (Haryana South, Western UP)	Wheat (Oct-May) → Mustard (Sept-Feb) intercropped; minimal overlap	Reduces water requirement by 20-30%; better suited to limited irrigation
Wheat-Pulses (Chickpea/Lentil) Rotation	Low-rainfall, light-soil zones (Rajasthan border regions)	Wheat (Oct-May) → Chickpea/Lentil (Oct-April); alternate years with fallow	Fixes nitrogen; reduces fertilizer input; improves soil structure
Sugarcane-Wheat-Mustard	High-value crop regions (Punjab, UP sugarbelt)	Sugarcane (11-month) + Wheat intercrop (Oct-May) → Mustard (Sept-Feb)	Sugarcane provides perennial income; intercropping optimizes land use

5.2 Varietal Recommendations (Climate-Resilient Varieties - ICAR Approved 2025-26)

Wheat (Rabi):

- **High-yield + Yellow Rust Resistant:** PBW 343, PBW 502, HD 3118, WH 1105 (all carry rust-resistance genes Yr24/Yr25)
- **Heat-tolerant (for late-sown/terminal heatwave risk):** PBW 644, DBW 187 (better grain-filling under heat stress)

Rice (Kharif):

- **Stem Borer-Resistant:** IR 36, ADT 47, Sugandhya (inherent resistance to SB)
- **Flood-tolerant:** Saryu, Sukhamali (for flash-flood prone areas)

Mustard (Rabi):

- **Aphid-Resistant/Tolerant:** Pusa Mustard 27, PM 30, RH 0406, NRCHB 101 (poor host for mustard aphid)
- **High-yield:** Pusa Mustard 25, RLC 3 (yield: 25-28 q/ha under optimal conditions)

Section 6: Implementation Roadmap & Farmer Action Calendar

Wheat (Rabi 2025-26)

Month	Activity	Responsible Actor	Expected Output
Aug-Sept	Soil sampling for SHC (if last SHC >3 years old); FYM/compost application (5 t/ha)	Farmer + Village Sampling Team	SHC report by Oct-end; field preparation complete
October	Seed procurement (certified seed, resistant variety); field clearance; seeding rate calibration (100 kg/ha)	Farmer + Seed Dealer	Disease-free seeds secured; field ready for sowing
Nov 1-15	Optimal sowing window; irrigation scheduling (CRI at 21 DAS)	Farmer	Crop establishment; good tillering by December
December	First nitrogen top-dressing (50 kg N/ha); field monitoring for yellow rust appearance	Farmer + VEW	Robust vegetative growth; disease surveillance initiated
January-February	Second irrigation (tillering); monitor yellow rust; initiate spray if pustules detected on lower leaves	Farmer + Pest Monitor	Controlled disease progression; early fungicide application if needed
March	Boot-stage irrigation; boot-stage fungicide application (if rust incidence >5%); heatwave contingency prep (anti-transpirants, mulch readiness)	Farmer	Grain-filling phase; heat-stress mitigation in place
April-May	Harvest readiness; monitor grain fill; accelerate harvest if unseasonal rain forecast; threshing floor preparation	Farmer + Labor	Timely harvest; minimize shattering & sprouting losses

Rice (Kharif 2025-26)

Month	Activity	Responsible Actor	Expected Output
April-May	Nursery preparation (water, seed treatment with Carbendazim @ 2.5 g/kg seed); SHC implementation (nutrient recommendations); field leveling	Farmer + Nursery Manager	Healthy seedlings; field drainage optimized
May-June	Optimal sowing window; transplanting at 30-35 DAS; biofertilizer (Azospirillum) application; pheromone trap setup (3-5 traps/acre)	Farmer	Uniform transplant; biological monitoring initiated
July (Peak Monsoon)	Active water management (5-7 cm standing water); stem borer surveillance via pheromone traps; Trichogramma release (50,000-100,000 wasps/ha)	Farmer + Entomologist	Controlled water level; parasitoid establishment
Late July-August	Panicle initiation (50-60 DAS); first insecticide spray if stem borer captures >8 moths/trap/night; second Trichogramma release	Farmer	Deadheart prevention; biological control reinforcement
August-September	Heading (80-95 DAS); continue water management; fungicide spray if blast symptoms appear (Tebuconazole 0.1%)	Farmer	Panicle exertion; disease suppression
September-October	Pre-harvest assessment; drain field 10-15 days before harvest; monitor for unseasonal rain risk; prepare harvesting equipment	Farmer + Labor	Grain maturation; harvest readiness

Mustard (Rabi 2025-26)

Month	Activity	Responsible Actor	Expected Output
August-September	Soil sampling (SHC if due); field preparation; resistant variety selection (Pusa Mustard 27, PM 30); seed procurement	Farm er	SHC report by Sept 30; disease-free seeds ready
Sept 25-Oct 20	Optimal sowing window ; seed treatment (Carbendazim @ 2.5 g/kg); sowing at 4-5 kg seed/ha	Farm er	Uniform germination; no seed-borne diseases
November	Seedling thinning (spacing: 20-25 cm); first irrigation if rainfall <40 mm in October	Farm er	Optimized plant density; root establishment
December	Yellow sticky trap deployment (4-5 traps/acre); flower bud stage monitoring for aphid incidence	Farm er + Pest Scout	Aphid population baseline established; traps attracting pests
January	Peak aphid infestation period ; initiate NSKE/neem oil spray (100-150 aphids/plant threshold); continue monitoring	Farm er	Organic pest control active; sprays at economic threshold
Late January-February	Early pod stage; decision on chemical spray (if aphid >150/plant, use Thiamethoxam 0.2 g/L); maximize spray coverage on leaf undersides	Farm er	IPM decision-making; chemical applied only if organic failed

Month	Activity	Responsible Actor	Expected Output
February-March	Seed fill stage; reduce irrigation (maintain drying); pre-harvest assessment; arrange harvesting labor	Farm er	Crop maturation; harvest schedule confirmed
March-April	Harvest (when pods turn yellowish-brown); threshing; grain drying to <8% moisture; storage in airtight containers	Farm er	Quality seeds; ready for market sale or next-season planting

Glossary of Technical Terms

Term	Definition	North Indian Context
DAS	Days After Sowing	Standard agronomic measurement in crop development
Tillering	Emergence of secondary shoots from leaf axils	Critical stage for nitrogen application (30-40 DAS in wheat)
Panicle Initiation (PI)	Transition from vegetative to reproductive growth; visible in rice at 50-60 DAS	Timing for potassium application & pest management
Boot Stage	Pre-flowering; spike/panicle emergence from leaf sheath	Fungicide application stage for yellow rust
Deadhearts (DH)	Yellowing/drying of central leaf in rice due to stem borer larval damage	Economic Threshold: 5-10% DH
Whiteheads (WH)	Unfilled panicles in rice due to stem borer damage at PI	Visible at 100+ DAS; grain loss >80% if severe
Uredia	Pustule-like fruiting bodies of fungal pathogens; contain spores	Yellow rust pustules: 1-2 mm linear streaks on wheat leaves
Fusarium Head Blight (FHB)	Fungal disease of wheat; pre-harvest sprouting risk; toxin concern	Controlled via resistant varieties + crop rotation
PMFBY	Pradhan Mantri Fasal Bima Yojana (crop insurance scheme)	Premium: 2% (kharif), 1.5% (rabi); government bears 88%
SHC	Soil Health Card	Issued once per 3 years; tests 12 soil parameters
Economic Threshold	Pest population level at which management cost =	Varies by crop (wheat: 5% yellow rust; rice: 5-

Term	Definition	North Indian Context
(ET)	crop loss prevention benefit	10% DH)

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- **Vikaspedia package of practices** (Ministry of Agriculture, GoI)
- **Farmer feedback from demonstration villages** (North Indian Plains KVKs)
- **Policy framework:** PM-KISAN (2024-26) & Soil Health Card Scheme (2015-2026)

Final Notes for Extension Workers & Farmers

1. **Customization:** Recommendations are standardized for North Plains; local micro-variations (field elevation, soil type, water source) may require site-specific fine-tuning via VEW/KVK consultation.
2. **Cost-Benefit:** Adoption of climate-resilient practices (SHC + resistant varieties + IPM) reduces input cost by 15-20% while increasing yield by 10-15%; ROI typically 200-300% within single season.
3. **Knowledge Dissemination:** Share this advisory via **ATMA meetings, KVK training, and WhatsApp farmer groups** for rapid uptake.

4. **Feedback Mechanism:** Farmers encouraged to report field outcomes (yield, pest/disease severity, weather events) to ATMA/KVK for continuous refinement of advisory.

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